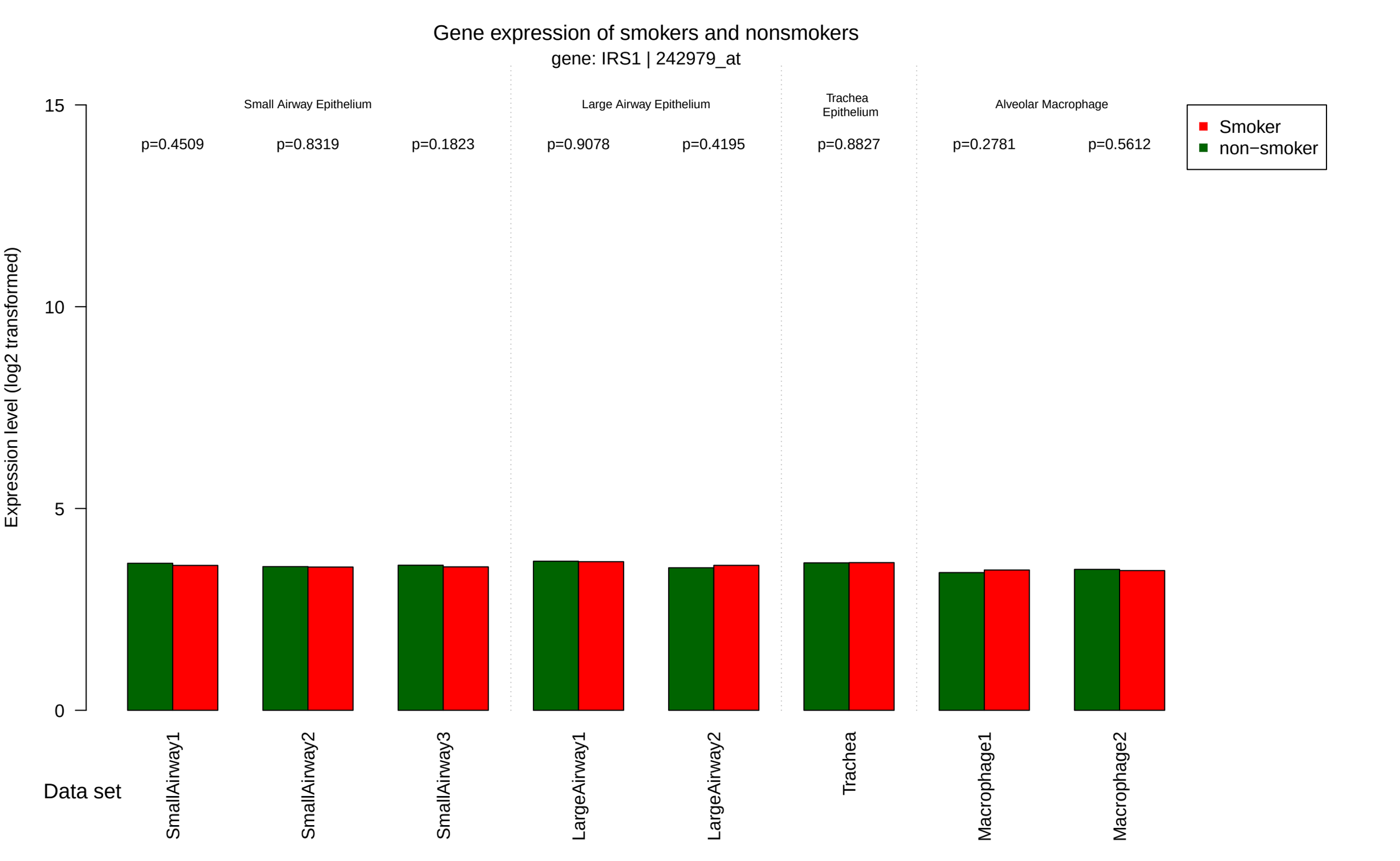
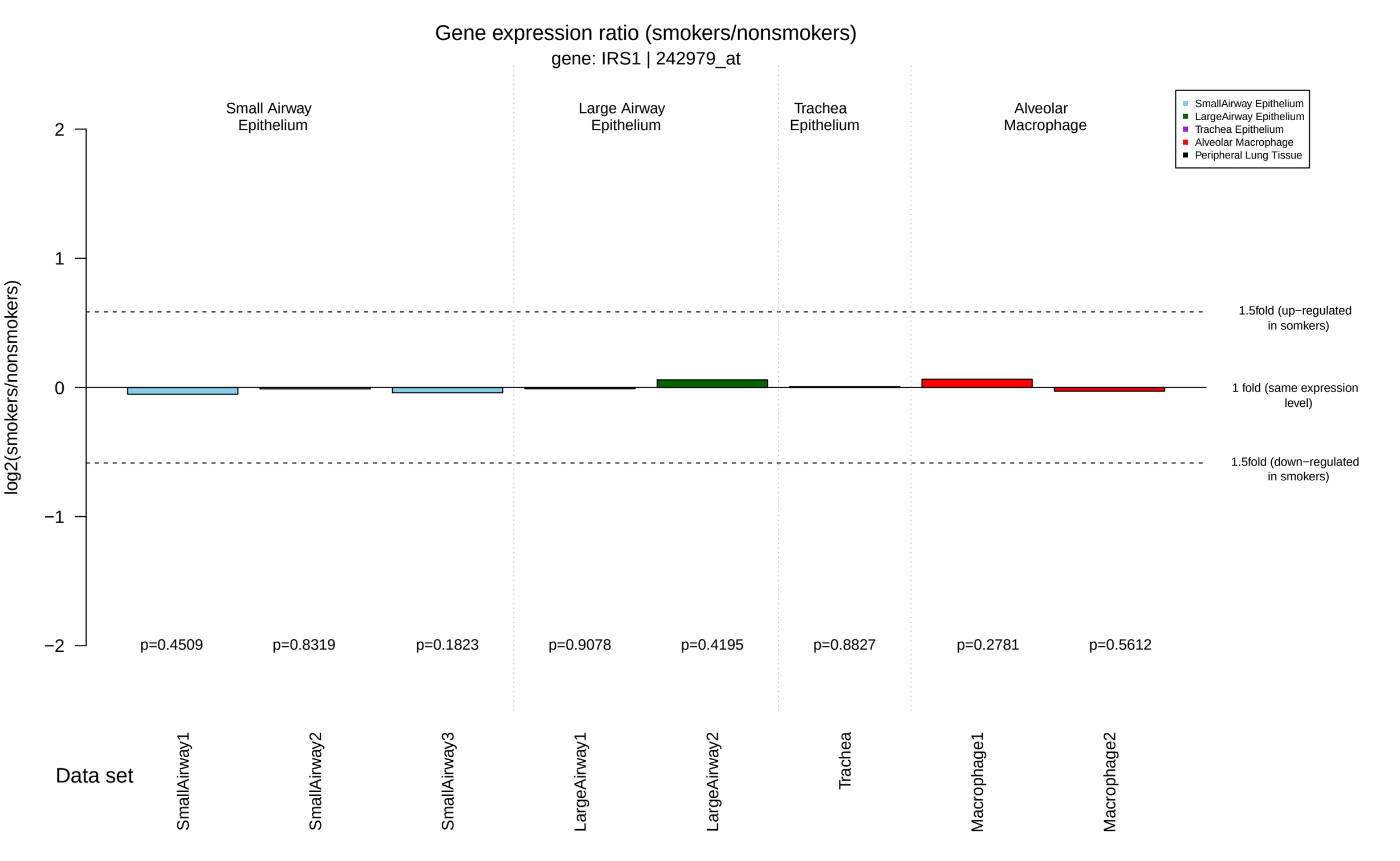
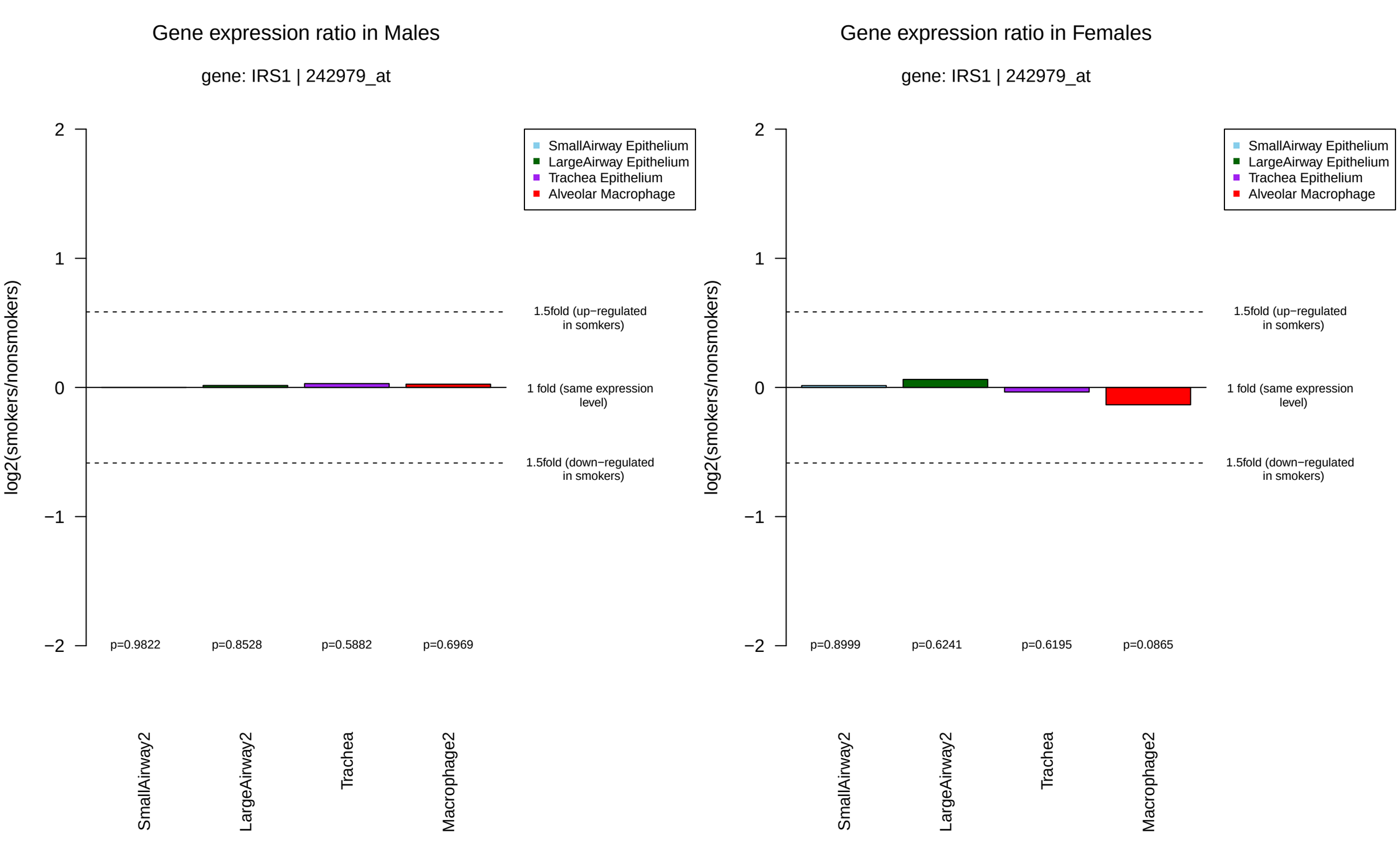
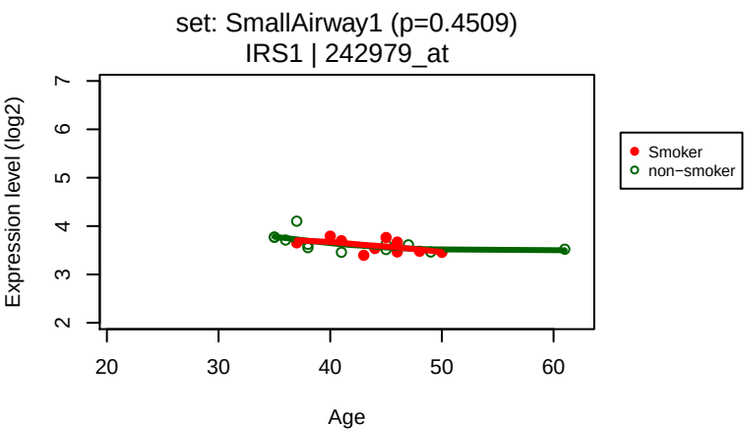
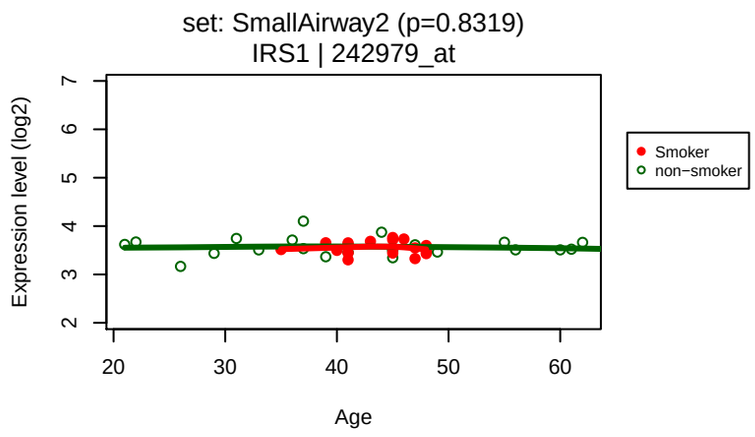
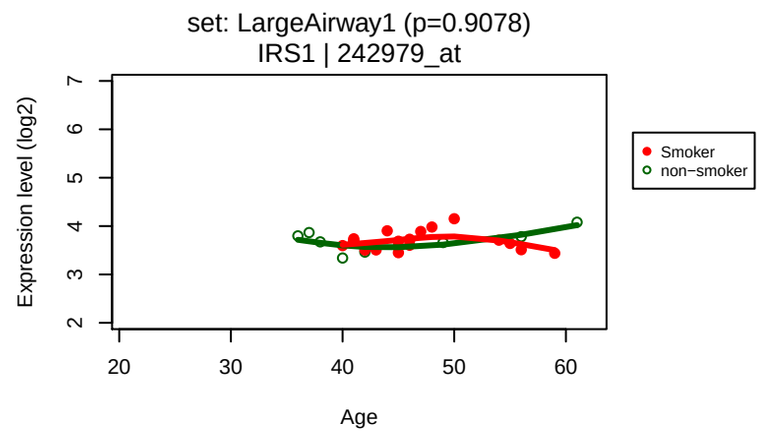

Gene expression of smokers and nonsmokers
gene: IRS1 | 242979_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4509
p=0.8319
p=0.1823
p=0.9078
p=0.4195
p=0.8827
p=0.2781
p=0.5612
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

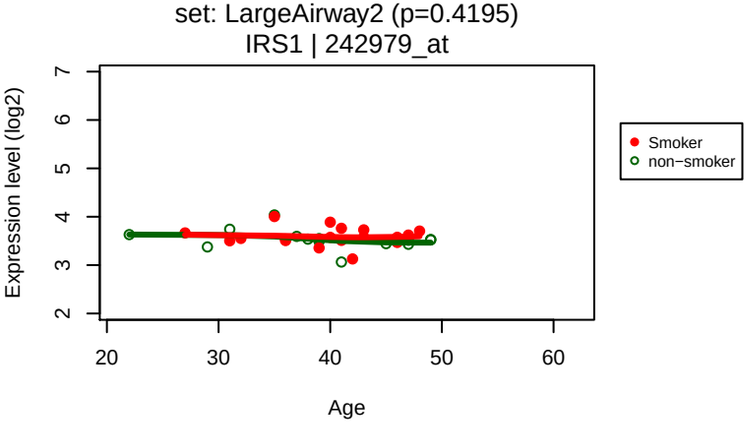

Gene expression ratio (smokers/nonsmokers)
gene: IRS1 | 242979_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4509
p=0.8319
p=0.1823
p=0.9078
p=0.4195
p=0.8827
p=0.2781
p=0.5612
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

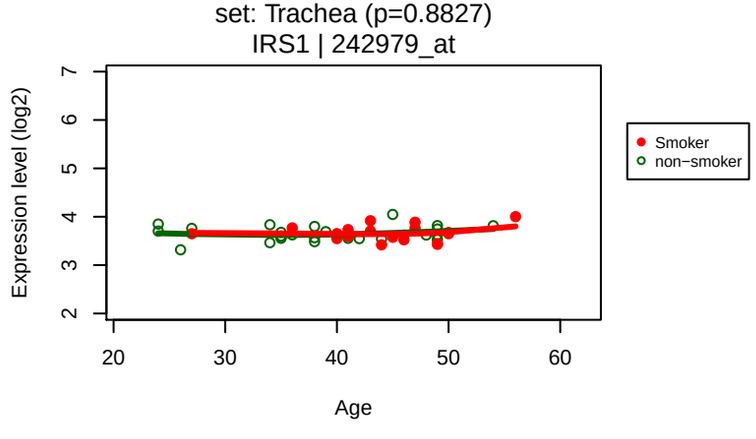

Gene expression ratio in Males
gene: IRS1 | 242979_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9822
p=0.8528
p=0.5882
p=0.6969
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRS1 | 242979_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8999
p=0.6241
p=0.6195
p=0.0865
SmallAirway2
LargeAirway2
Trachea
Macrophage2

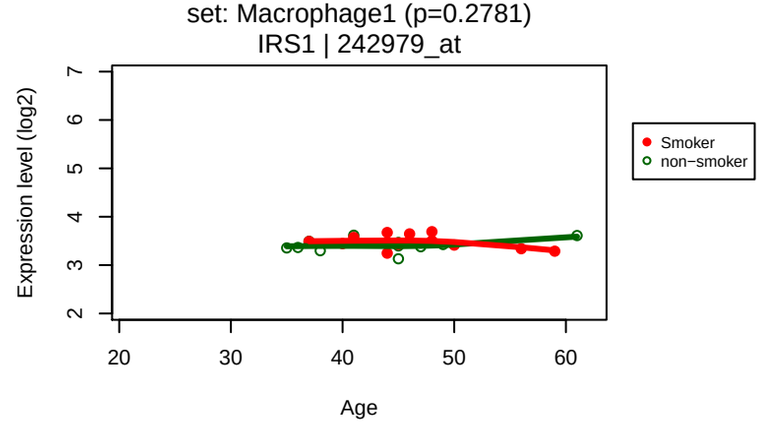

set: SmallAirway1 (p=0.4509)
IRS1 | 242979_at
Expression level (log2)
Age
Smoker
non−smoker

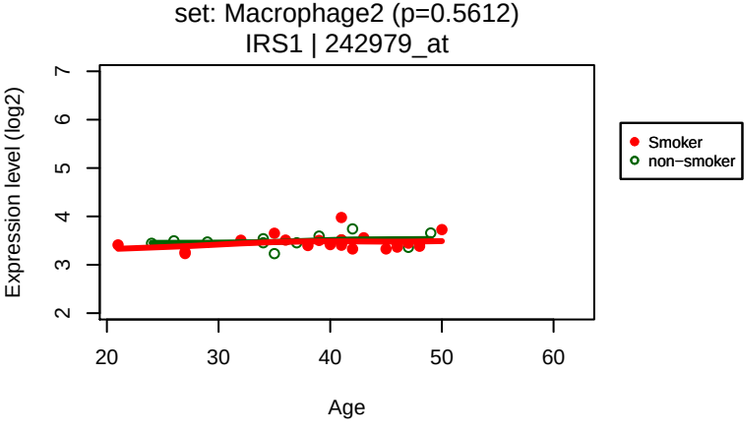

set: SmallAirway2 (p=0.8319)
IRS1 | 242979_at
Expression level (log2)
Age
Smoker
non−smoker

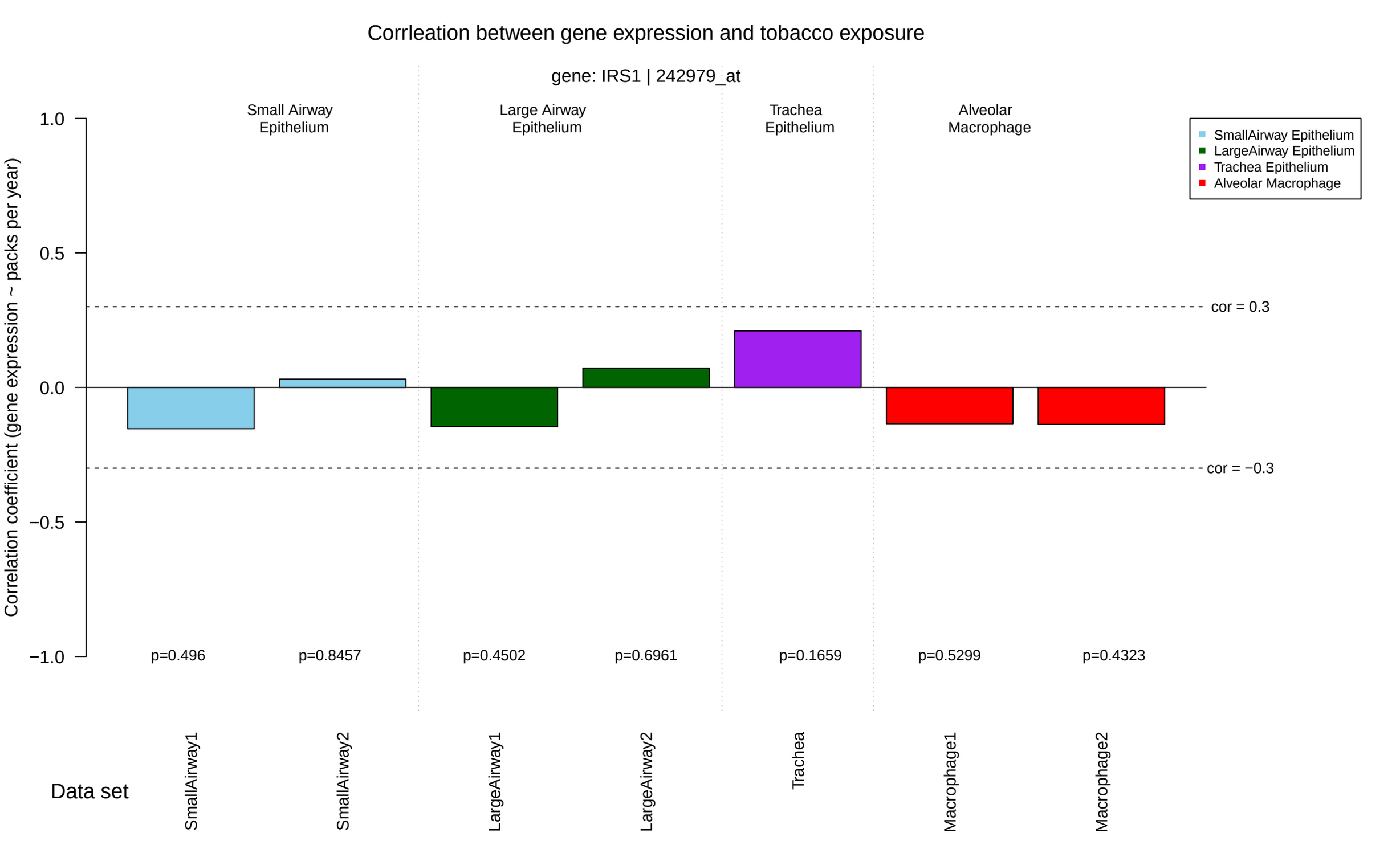

set: LargeAirway1 (p=0.9078)
IRS1 | 242979_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4195)
IRS1 | 242979_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8827)
IRS1 | 242979_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2781)
IRS1 | 242979_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5612)
IRS1 | 242979_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRS1 | 242979_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.496
p=0.8457
p=0.4502
p=0.6961
p=0.1659
p=0.5299
p=0.4323
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2